Preparation of a Buffer with a Given pH

In experimental work, a chemist or biologist usually first decides the pH at which a particular system should be buffered and then chooses the buffer components that will best deliver this pH. This choice requires careful attention to what most affects the pH of a buffer.

The Factors That Govern the pH of a Buffer Solution

We can more clearly see the two factors that dominate the pH of a buffered solution by rearranging Equation.12 to solve for $[H^+]$.

$$[H^+] = K_a \frac{[HA]}{[A^-]} \qquad (13)$$

or

$$[H^+] = K_a \frac{\text{mol } HA}{\text{mol } A^-} \qquad (14)$$

The first factor to affect $[H^+]$ (and therefore pH) is the K_a of the weak acid. The second is the ratio of the molarities (in Equation.13) or the ratio of moles (in Equation.14) . The quantities we use in these two equations, of course, can be the initial values—either concentrations or moles. So let's rewrite them as follows :

$$[H^+] = K_a \frac{[HA]_{initial}}{[A^-]_{initial}} \qquad (15)$$

$$[H^+] = K_a \frac{(\text{mol } HA)_{initial}}{(\text{mol } A^-)_{initial}} \qquad (16)$$

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Notice particularly what happens if we prepare a buffer so as to make the concentrations of the two buffer components identical. Then the ratio $[HA]_{initial}/[A^-]_{initial}$ in Equation.15 equals 1, which means that $[H^+] = K_a$ and therefore $pH = pK_a$.

Selecting the Weak Acid for the Preparation of a Buffer Solution

Usuaily buffers are made so that the ratio $[HA]_{initial}/[A^-]_{initial}$ is not greatly different from one. Consequently, what mostly determines where, on the pH scale, a buffer can work best is the pK_a of the weak acid. Thus to prepare a specific buffer for use at a prechosen pH, we first select a weak acid whose pK_a is near the pH we desire. Almost never can an acid be found, however, whose pK, exactly equals the pH we want. So we try for an acid whose pK_a is close to this pH. Then, by experimentally adjusting the ratio $[HA]_{initial}/[A^-]_{initial}$, we can make a final adjustment to get the desired pH.

The desirable range for the ratio $[HA]_{initial}/[A^-]_{initial}$ is from 10/1 to 1/10. Outside this range we can run into problems with the solubilities of the solutes, or we can have a buffer of low capacity. (The question of buffer capacity will be studied soon.) The log of 10/1 is 1, and the log of 1/10 is -1. Therefore, the range for the ratio $[HA]_{initial}/[A^-]_{initial}$ of 10/1 to 1/10 works out to the desirable range of pH values for a buffer normally being

$$pH = pK_a \pm 1 \qquad (17)$$

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EX.15

A solution buffered at pH 5 is needed in a chemistry experiment. Can we use acetic acid and sodium acetate to make it? If so, what ratio of acetic acid to acetate Preparing a Buffer Solution to ion is needed?

Analysis

We first check the pK_a of acetic acid to see if it is in the desired range of $pH = pK_a \pm 1$. If it is, then we proceed to calculate the ratio.

Solution

Because we want the pH to be 5, the pK_a of the selected acid should be 5 ± 1, meaning the range of pK_a between 4 and 6. Because $K_a = 1.8 \times 10^{-5}$ for acetic acid, $pK_a = 4.74$. So the pK_a of acetic acid falls in the desired range, and acetic acid can be used together with the acetate ion to make the buffer.

We will use Equation.16 to answer the second question: What mole ratio of solutes is needed?

$$[H^+] = K_a \frac{(\text{mol } HC_2H_3O_2)_{\text{initial}}}{(\text{mol } C_2H_3O_2^-)_{\text{initial}}}$$

Solving for the mole ratio gives:

$$\frac{(\text{mol } HC_2H_3O_2)_{\text{initial}}}{(\text{mol } C_2H_3O_2^-)_{\text{initial}}} = \frac{[H^+]}{K_a}$$

The desired pH 5.00,so $[H^+]=1.0 \times 10^{-5}$ also $K_a = 1.8 \times 10^{-5}$. Substituting gives:

$$\frac{(\text{mol } HC_2H_3O_2)_{\text{initial}}}{(\text{mol } C_2H_3O_2^-)_{\text{initial}}} = \frac{1.0 \times 10^{-5}}{1.8 \times 10^{-5}} = 0.56$$

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This is the mole ratio of the buffer components we want. So we have to prepare the solution such that

$$\text{mol } HC_2H_3O_2 = 0.56 \times \text{mol } C_2H_3O_2^-$$

Suppose that we want 1 L of the solution, and that we decide to use sodium acetate as the source of the acetate ion. We could, for example, dissolve 0.1 mol of $NaC_2H_3O_2$ in water and then add 0.056 mol of $HC_2H_3O_2$, making the final volume equal to 1 L. We could have chosen quantities at one-tenth this, for example, 0.01 mol of $NaC_2H_3O_2$ and 0.0056 mol of $HC_2H_3O_2$. The ratio would still be the same. The solution would still be buffered at a pH of 5.

Buffer Capacity

Earlier we hinted at the question of the capacity of a buffer—**how much extra acid or base the buffer solution can absorb before its buffering ability is essentially destroyed.** It would always be possible to add so much strong acid to a buffered solution that all the base present would be neutralized. Or all of the buffer's acid could be destroyed by adding too much strong base. **The ratio of anion to acid influences only the pH of the buffered solution, not its capacity.**

The buffer's capacity is determined by the sizes of the actual molarities of these components. So the chemist must decide before making the buffer solution what outer limits to the change in pH can be tolerated—0.1 unit, 0.2 unit, or something else. Then, consistent with the necessary ratio of acid to anion, the chemist calculates the actual number of moles of each to dissolve in the solution. Let's do a calculation to show how even a relatively small amount of extra acid (or base) can severely tax a buffer's capacity.

EX.16

A student prepared 1L of an acetic acid—sodium acetate buffer that included 1M $HC_2H_3O_2$ and 1M $NaC_2H_3O_2$. This entire solution was intended for experiment in which 0.15 mol of H^+ was expected to be generated by a reaction without

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changing the volume of the solution. Is there enough buffer capacity to neutralize this much strong acid without letting the pH change by more than 0.1unit?

Analysis

To answer the question, we need to know the initial pH of the buffer and the pH after the addition of the H_3O^+. We use the given concentrations of the $HC_2H_3O_2$ and $C_2H_3O_2^-$ to obtain the initial pH. Next, we determine how the concentrations of the $HC_2H_3O_2$ and $C_2H_3O_2^-$ change when the system absorbs the strong acid. We then use these new concentrations to calculate the new pH of the buffer. The difference between the new and original pH's tells us how much the pH changes.

Solution

First, we calculate the pH of the buffer before addition of the H_3O^+ For acetic acid, we have seen that:

$$K_a = \frac{[H^+][C_2H_3O_2^-]}{[HC_2H_3O_2]} = 1.8 \times 10^{-5}$$

The given data tell us that initially,

$$[HC_2H_3O_2]_{initial} = 1M$$

$$[C_2H_3O_2^-]_{initial} = 1M$$

Substituting these values lets us calculate the initial pH of the buffer.

$$\frac{[H^+](1.00)}{1.00} = 1.8 \times 10^{-5}$$

$$[H^+] = 1.8 \times 10^{-5}$$

$$pH = 4.74$$

The H^+ formed in the solution reacts with acetate as follows:

$$H^+ + C_2H_3O_2^- \longrightarrow HC_2H_3O_2$$

The reaction is nearly complete, in the sense that virtually all the added H^+ is consumed. As an approximation, we assume complete reaction, so for each mole of

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H^+ added, a mole of $C_2H_3O_2^-$ is changed to a mole of $HC_2H_3O_2$. Since 0.15 mol of H^+ is added,

$$[HC_2H_3O_2]_{initial} = (1.00 + 0.15)M = 1.15\ M$$

$$[C_2H_3O_2^-]_{initial} = (1.00 - 0.15)M = 0.85\ M$$

We now substitute these values into the mass action expression to calculate the final $[H^+]$ in the solution.

$$\frac{[H^+](0.85)}{(1.15)} = 1.8 \times 10^{-5}$$

$$[H^+] = 2.4 \times 10^{-5}\ mol\ L^{-1}$$

From this we calculate the final pH,

$$pH = 4.62$$

By adding 0.15 mol of HCl to the buffer, its pH has gone from 4.74 to 4.62, a change greater than the limit of ± 0.1 pH unit. The buffer solution, in other words, could not absorb even 0.15 mol of H^+ without exceeding the limit, even though it began with 1.00 mol of the base, $C_2H_3O_2^-$.

The Effectiveness of a Buffer System

Although buffers do not have unlimited capacities, they still operate remarkably well in preventing wide swings in pH. The question of capacity is related to the question of buffer effectiveness. How well does a buffer system work? How well does it actually hold the pH? In Example.20, we added 0.15 mol of strong acid to 1.00 L of a buffer and the pH changed by 0.12 pH unit. Just think of how much the pH would change had we added this much acid to 1.00 L of pure water. The solution would now have

$$[H^+] = 0.15\ mol\ L^{-1} \quad \text{and} \quad pH = -\log(0.15) = 0.82$$

With the buffer, the pH changes from 4.74 to 4.62; without the buffer the pH (starting with pure water) changes from 7.00 to 0.82. This is an enormous difference.

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8. Acid-Base Titrations

Ideally, the end point should occur at the equivalence point, when equivalent amounts of acid and base have combined. We will understand this better by studying how the pH of the solution slowly added from a buret to a so- being titrated changes with the addition of titrant.

When the pH of a solution at different stages of a titration is plotted against the volume of titrant added, we obtain a titration curve. The values of pH in these plots can be calculated by the procedures studied in this chapter, so this discussion will serve as a review.

Titration of a Strong Acid by a Strong Base

The titration of HCl(aq) with standard NaOH(aq) illustrates the titration of a strong acid by a strong base. The molecular and net ionic equations are:

$$HC1(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O$$

$$H^+(aq) + OH^-(aq) \rightarrow H_2O$$

Let's consider what happens to the pH of a solution, initially 25 mL of 0.2M HC1, as small amounts of the titrant, 0.2 M NaOH, are added. We will calculate the pH of the resulting solution at various stages of the titration, retaining only two significant figures, and plot the values against volume titrant.

Before any titrant has been run into the receiving flask, the pH of the HC1 solution in this flask is found as follows. Because HC1 is a strong acid, we know that

$$[H^+] = [HC1] = 0.2M$$

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So **pH = - log(0.2) = 0.7 (the initial pH)**

The amount of HC1 initially present in 25 mL of 0.2M HC1 is:

$$25.00 \text{ mL HCl soln} \times \frac{0.2000 \text{ mol HCl}}{1000 \text{ mL HCl soln}} = 5.000 \times 10^{-3} \text{ mol HCl}$$

Now suppose we add 10.00 mL of 0.2000 M NaOH from the buret. The amount of NaOH added then is:

$$10.00 \text{ mL NaOH soln} \times \frac{0.2000 \text{ mol NaOH}}{1000 \text{ mL NaOH soln}} = 2.000 \times 10^{-3} \text{ mol NaOH}$$

This base neutralizes some of the acid, with one H^+ ion removed for each OH^- ion added, as the net ionic equation shows. We calculate the amount of HC1 left, therefore, by simple subtraction:

$(5 \times 10^{-3} - 2 \times 10^{-3})$ mol HCl = 3×10^{-3} mol HCl remaining

The total volume of the solution is now:

(25 + 10) mL=35mL = 0.035 L, so the $[H^+]$ is calculated as:

$$[H^+] = \frac{3.000 \times 10^{-3} \text{ mol}}{0.03500 \text{ L}} = 8.571 \times 10^{-2} M$$

The corresponding pH is 1.07.

The following table shows the results of the calculations of the pH of the solution in the receiving flask after the addition of further small volumes of the titrant. After 25 mL of 0.2 M NaOH is added, all of the HC1 is neutralized. At this point the solution is a mixture of Na^+ and Cl^- ions, neither of which have an effect on pH, so the solution is neutral with a pH 7.

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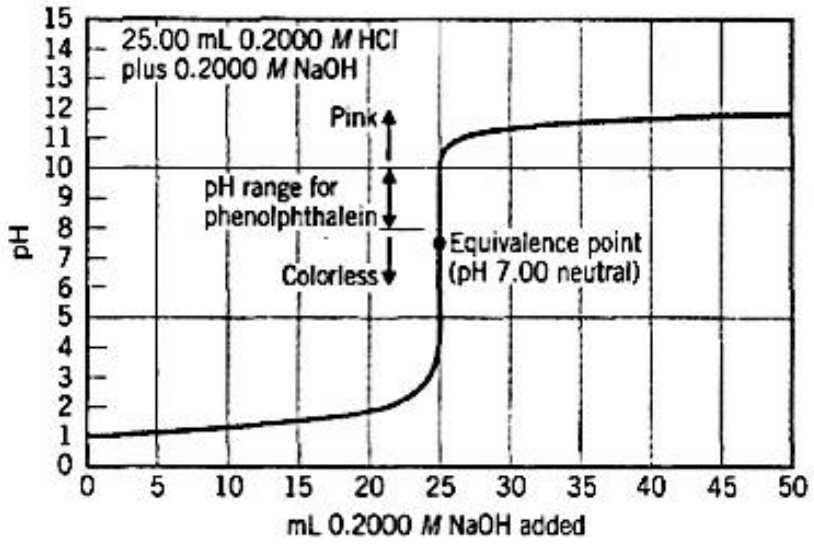
Titration of 25.00 mL of 0.2000 *M* HCl with 0.2000 *M* NaOH

Initial Volume of HCl mL)	Initial Amount of HCl (mol)	Volume of NaOH Added (mL)	Amount of NaOH (mol)	Amount of Excess Reagent (mol)	Total Volume of Solution (mL)	Molarity of Ion in Excess (mol L^{-1})	pH
25.00	5.000×10^{-3}	0	0	$5.000 \times 10^{-3}(H^+)$	25.00	$0.2000(H^+)$	0.70
25.00	5.000×10^{-3}	10.00	2.000×10^{-3}	$3.000 \times 10^{-3}(H^+)$	35.00	$8.571 \times 10^{-2}(H^+)$	1.07
25.00	5.000×10^{-3}	20.00	4.000×10^{-3}	$1.000 \times 10^{-3}(H^+)$	45.00	$2.222 \times 10^{-2}(H^+)$	1.65
25.00	5.000×10^{-3}	24.00	4.800×10^{-3}	$2.000 \times 10^{-4}(H^+)$	49.00	$4.082 \times 10^{-3}(H^+)$	2.39
25.00	5.000×10^{-3}	24.90	4.980×10^{-3}	$2.000 \times 10^{-5}(H^+)$	49.90	$4.000 \times 10^{-4}(H^+)$	3.40
25.00	5.000×10^{-3}	24.99	4.998×10^{-3}	$2.000 \times 10^{-6}(H^+)$	49.99	$4.000 \times 10^{-5}(H^+)$	4.40
25.00	5.000×10^{-3}	25.00	5.000×10^{-3}	0	50.00	0	7.00
25.00	5.000×10^{-3}	25.01	5.002×10^{-3}	$2.000 \times 10^{-6}(OH^-)$	50.01	$3.999 \times 10^{-5}(OH^-)$	9.60
25.00	5.000×10^{-3}	25.10	5.020×10^{-3}	$2.000 \times 10^{-5}(OH^-)$	50.10	$3.992 \times 10^{-4}(OH^-)$	10.60
25.00	5.000×10^{-3}	26.00	5.200×10^{-3}	$2.000 \times 10^{-4}(OH^-)$	51.00	$3.922 \times 10^{-3}(OH^-)$	11.59
25.00	5.000×10^{-3}	50.00	1.000×10^{-2}	$5.000 \times 10^{-3}(OH^-)$	75.00	$6.666 \times 10^{-2}(OH^-)$	12.82



Titration curve for titrating a strong acid (0.2000 M HCI with a strong base (0.2 M NaOH).

Any further addition of titrant is simply the addition of more and n ore OH^- ion to (an increasing volume of) the solution in the flask. So after 25.00 mL of NaOH solution has been added, the pH calculation reflects only a dilute solution of NaOH.

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The figure shows a plot of the pH of the solution in the receiving flask versus the milliliters of the 0.2 M NaOH solution added—the titration curve for the titration of a strong acid with a strong base. Notice how slowly the pH of the solution increases until we are almost exactly at the equivalence point. pH 7. With the addition of more base, the curve continues to rise very sharply and then, almost as suddenly, bends to reflect a very gradual increase in pH. Notice also that the equivalence point occurs at a pH of 7, which characterizes the titration of any strong monoprotic acid with any strong base. The only ions at the equivalence point in our example, aside from the traces of those from the self-ionization of water, are Na^+ and Cl^-. Na^+ is not an acidic cation and Cl^- is not a basic anion, so the solution at this point is neutral with a pH of 7.

Titration of a Weak Acid by a Strong Base

Let us do the calculations and draw a titration curve for the titration of 0.2 M $HC_2H_3O_2$ with 0.2 M NaOH. The molecular and net ionic equations are:

$$HC_2H_3O_2\,(aq) + NaOH(aq) \rightarrow NaC_2H_3O_2\,(aq) + H_2O$$

$$HC_2H_3O_2\,(aq) + OH^-\,(aq) \rightarrow C_2H_3O_2^-\,(aq) + H_2O$$

Let us start with 25 mL of 0.2 M $HC_2H_3O_2$ and use 0.2 M NaOH as the titrant. Because the concentrations of both solutions are the same, we know that we will need exactly 25 mL of the base to reach the equivalence point. With this as background, let's look at what is involved in calculating the pH at various points along the titration curve.

1. Before the titration begins.

The solution at this point is simply a solution of the weak acid, $HC_2H_3O_2$. We must use K_a to calculate the pH.

2. During the titration but before the equivalence point.

As we add NaOH to the $HC_2H_3O_2$ the neutralization reaction produces $C_2H_3O_2^-$, so the solution contains both $HC_2H_3O_2$ and $C_2H_3O_2^-$ (it is a buffer solution). We can use K to calculate the pH here as well.

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3. At the equivalence point.

Here all the $HC_2H_3O_2$ has reacted and the solution contains the salt $NaC_2H_3O_2$ (i.e., we have a mixture of the ions Na^+ and $C_2H_3O_2^-$) . The anion is a weak base, so we use K_b to calculate the pOH and then the pH.

4. After the equivalence point.

The OH^- introduced by the further addition of NaOH has nothing with which to react and causes the solution to become basic. The concentration of OH^- produced by the unreacted base is used to calculate the pOH and ultimately the pH. We will look at each of these briefly.

Before the Titration Begins

The question here is, What is the pH of 0.2M $HC_2H_3O_2$?

For acetic acid, $K_a = 1.8 \times 10^{-5}$, we let $x = [H^+] = [C_2H_3O_2^-]$, and use $[HC_2H_3O_2] = 0.2$ M.

$$K_a = \frac{[H^+][C_2H_3O_2^-]}{[HC_2H_3O_2]} = \frac{(x)(x)}{(0.2000)} = 1.8 \times 10^{-5}$$

$$x = 1.9 \times 10^{-3}$$

So, $[H^+] = 1.9 \times 10^{-3}$M, which yields a pH of 2.72.This is the pH before any NaOH is added.

During the Titration but Before the Equivalence Point

Once we add NaOH, the solution contains both $HC_2H_3O_2$ and $C_2H_3O_2^-$, so we now have a buffer system. Let's calculate the pH after 10.00 mL of the base has been added. The amount of NaOH added is:

$$10.00 \text{ mL NaOH soln} \times \frac{0.2000 \text{ mol NaOH}}{1000 \text{ mL soln}} = 2.000 \times 10^{-3} \text{ mol NaOH}$$

We have thereby neutralized 2×10^{-3} mol of $HC_2H_3O_2$, and we have produced 2×10^{-3} mol of $C_2H_3O_2^-$. The amount of $HC_2H_3O_2$ remaining in the solution (realizing we began with 5×10^{-3} mol), is simply

$$(5 \times 10^{-3} - 2 \times 10^{-3}) \text{ mol } HC_2H_3O_2 = 3 \times 10^{-3} \text{ mol } HC_2H_3O_2$$

The total volume is now (25 mL + 10 mL) 35 mL 0.035 L.

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The concentrations of acetate ion and acetic acid now are, therefore,

$$[C_2H_3O_2^-] = \frac{2.000 \times 10^{-3}\ \text{mol}}{0.03500\ \text{L}} = 5.714 \times 10^{-2}\ \text{mol L}^{-1}$$

$$[HC_2H_3O_2] = \frac{3.000 \times 10^{-3}\ \text{mol}}{0.03500\ \text{L}} = 8.571 \times 10^{-2}\ \text{mol L}^{-1}$$

We can now substitute these values into the mass action expression and calculate $[H^+]$.

$$\frac{[H^+](5.714 \times 10^{-2})}{(8.571 \times 10^{-2})} = 1.8 \times 10^{-5}$$

Therefore,

$$[H^+] = 2.7 \times 10^{-5}\ \text{M} \qquad \text{pH} = 4.57$$

As long as we have not reached the equivalence point—as long as unreacted acetic acid remains together with acetate ion —we can compute the pH of the solution in the receiving flask by this kind of buffer calculation.

At the Equivalence Point

Once we exactly reach the equivalence point the solution is simply a dilute solution of sodium acetate. It no longer is a buffer. We now have to deal with the reaction of the acetate ion with water and the associated base ionization constant.

$$C_2H_3O_2^-\,(aq) + H_2O = HC_2H_3O_2\,(aq) + OH^-(aq)$$

$$K_b = \frac{[OH^-][HC_2H_3O_2]}{[C_2H_3O_2^-]} = \frac{K_w}{K_a} = 5.6 \times 10^{-10}$$

At the equivalence point, we have added 25 mL of 0.2 M NaOH to the original 25 mL of 0.2M $HC_2H_3O_2$. The volume is now 50 mL=0.05 L. We began with 5×10^{-3} mol of $HC_2H_3O_2$, so we now have 5×10^{-3} mol of $C_2H_3O_2^-$ (in 0.05 L). The concentration of $C_2H_3O_2^-$ therefore, is:

$$[C_2H_3O_2^-] = \frac{5.000 \times 10^{-3}\ \text{mol}}{0.05000\ \text{L}} = 0.1000\ M$$

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To solve the problem we realize that a small amount of the $C_2H_3O_2^-$ reacts to give equal amounts of OH^- and $HC_2H_3O_2$. We let the concentrations of the equal x, so $[OH^-] = [HC_2H_3O_2] = x$. Substituting into the mass action expression gives:

$$\frac{(x)(x)}{0.1000} = 5.6 \times 10^{-10}$$

$$x = 7.5 \times 10^{-6}$$

This means that $[OH^-] = 7.5\ X \times 10^{-6}$ M, so the pOH calculates to be 5.12. It is interesting that although we Finally, the pH is (14.00 - 5.12) or 8.88. Thus the pH at the equivalence point in this titration is 8.88, on the basic side because the solution contains the equivalence point the solution Bronsted base, $C_2H_3O_2$. In the titration of any weak acid with a strong base, the pH at the equivalence point will be greater than 7.

After the Equivalence Point

What happens if we continue to add NaOH(aq) to the solution beyond the equivalence point? Now the additional OH^- shifts the following equilibrium to the left.

$$C_2H_3O_2^-\ (aq) + H_2O = HC_2H_3O_2\ (aq) + OH^-\ (aq)$$

The generation of additional OH^- by this route is thus suppressed as we add more and more NaOH solution, so effectively the only source of OH^- to affect the pH is from the base added after the equivalence point. From here on. therefore, the pH calculations become identical to those for the last half of the titration curve for HCl and NaOH.Each point is just a matter of calculating the additional number of moles of NaOH, finding the new final volume, taking the ratio (the molar concentration), and then calculating pOH and pH.

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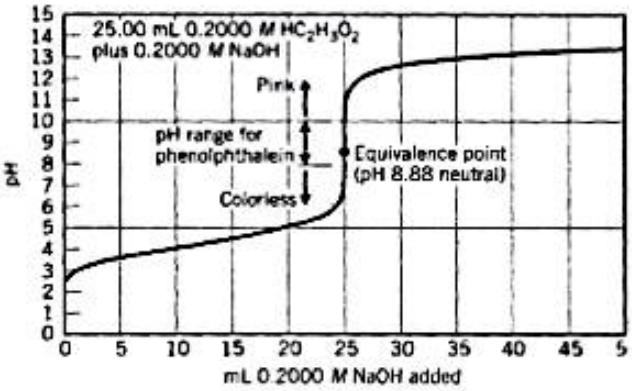
Titration of 25 mL of 0.2M $HC_2H_3O_2$ with 0.2M NaOH

Volume of Base Added (mL)	Molar Concentration of Species in Parentheses	pH
none	$1.9 \times 10^{-3}(H^+)$	2.72
10.00	$2.7 \times 10^{-5}(H^+)$	4.57
24.90	$7.2 \times 10^{-8}(H^+)$	7.14
24.99	$7.1 \times 10^{-9}(H^+)$	8.14
25.00	$7.5 \times 10^{-6}(OH^-)$	8.88
25.01	$4.0 \times 10^{-5}(OH^-)$	9.60
25.10	$4.0 \times 10^{-4}(OH^-)$	10.60
26.00	$3.9 \times 10^{-3}(OH^-)$	11.59
35.00	$3.3 \times 10^{-2}(OH^-)$	12.52



Titration curve for titrating a weak acid (0.2 M acetic acid)with a strong base (0.2 M NaOH).

Once again notice how the change in pH occurs slowly at first, then shoots up through the equivalence point, and finally looks just like the last half of the curve.

Titration of a Weak Base by a Strong Acid

The calculations involved here are nearly identical to those of the weak acid—strong base titration. We will review what is required but will not actually perform the calculations.

If we titrate 25 mL of 0.2M NH_3 with 0.2M HC1, the molecular and net ionic equations are:

$$NH_3(aq) + HC1(aq) \rightarrow NH_4C1(aq)$$

$$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$$

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1. Before the titration begins.

The solution at this point is simply a solution of the weak base, NH_3. Since the only solute is a base, we must use K_b to calculate the pH.

2. During the titration but before the equivalence point.

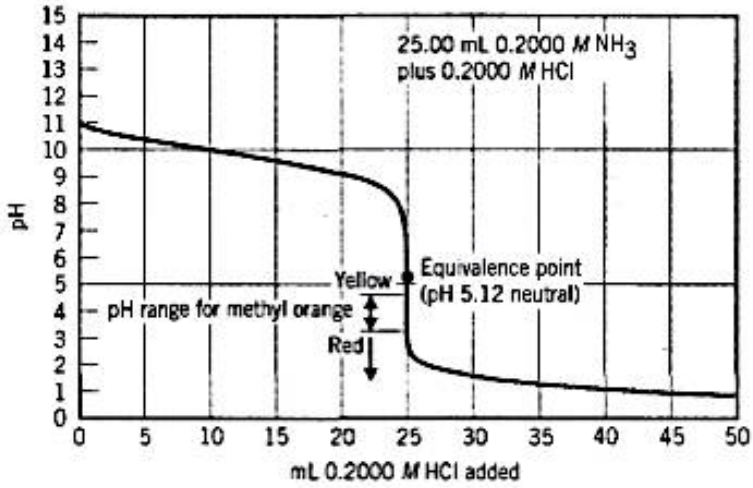
As we add HC1 to the NH_3 the neutralization reaction produces NH_4^+, so the solution contains both NH_3 and NH_4^+ (it is a buffer solution) . We can use K_b for NH_3 to calculate the pH here as well. (We could also use K_a if it were available.)

3. At the equivalence point.

Here all the NH_3 has reacted and the solution contains the salt NH_4Cl. The anion, $C1^-$, comes from the strong acid HC1 and is so weak a base that it cannot affect the pH. However, the cation NH_4^+ is a weak acid derived from the weak base NH_3 , so we use its Ka to calculate the pH.

4. After the equivalence point.

The H^+ introduced by further addition of HC1 has nothing with which to react and causes the solution to become acidic. The concentration of H^+ produced by the unreacted HC1 is used directly to calculate the pH.



Titration curve for the titration of a weak base (0.2M NH_3) with a strong acid (0.2M HC1).

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